

4.00 USD

1. Sentinel Smart Valve Actuator (Model S-WFC01)

- *Based on Ecowitt WFC01*
- Smart water timer valve designed to mitigate localized wildfire threat by auto-wetting property grounds when dry soil moisture thresholds trigger parametric alarm states.
- **Price:** \$79.00 USD

2. Automated Network Onboarding Workflow

The Shopify purchase acts as the initial transaction in the node onboarding lifecycle.

```
``mermaid
sequenceDiagram
    autonumber
    actor User as Policyholder / Buyer
    participant Shop as Shopify Storefront
    participant Webhook as Shopify Webhook Worker
    participant API as Chainmail API Server
    participant DB as Firestore Database
    participant Device as IoT Sentinel Device

    User->>Shop: Purchase Sentinel Kit (S-100 + S-700)
    Shop-->>User: Order Confirmation & Shipping Label Generated
    Shop->>Webhook: Event: order/paid (Payload contains email, address)
    Webhook->>API: POST /api/onboarding/provision
    activate API
```

```
API→API: Generate secure device key & pre-provision propertyId

API→DB: Write property & pre-provisioned sensors as PENDING_ACTIVATION

API-->>Webhook: Provisioning Success

deactivate API

Webhook-->>User: Onboarding Email with Device Credentials & Claim QR Code

User→Device: Mount hardware & connect to home Wi-Fi

Device→API: POST /api/ingest/ecowitt (First Telemetry Ping)

activate API

API→API: Authenticate device with secure pre-provisioned key

API→DB: Update status to VERIFIED_NODE & Activate Parametric Coverage

API-->>Device: Acknowledge (HTTP 200)

deactivate API

API→User: App Push Notification: "Node Active. Coverage Live!"

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```

3. API Provisioning Endpoint Specifications

When a checkout completes, Shopify sends an order/paid webhook to our backend API server:

Endpoint: **POST /api/onboarding/provision**

Access Control: HMAC Signature verified using Shopify API Webhook Shared Secret.

Request Payload (from Shopify):

```
`json
{
  "order_id": 987654321,
```

```
"email": "customer@example.com",

"shipping_address": {

  "address1": "12665 Julington Oaks Dr",

  "city": "Jacksonville",

  "province": "FL",

  "zip": "32223",

  "country": "US"

},

"line_items": [

{

  "sku": "SENTINEL-KIT-A",

  "quantity": 1,

  "name": "Sentinel Gateway & Wittboy Array"

}

]

}
```

Provisioning Logic:

1. **Address Validation:** Call Google Address Validation API to verify CASS status and retrieve exact rooftop coordinates (lat / lng).
2. **Database Record Creation:**
 - Create a new document in /properties:

```
`typescript
```

```
{
```

```
ownerEmail: "customer@example.com",

address: "12665 Julington Oaks Dr, Jacksonville, FL 32223",

lat: 30.1421936,

lng: -81.6157784,

riskScore: 72.4, // Calculated by Underwriting Agent based on coordinates

status: "PENDING_ACTIVATION",

createdAt: serverTimestamp()

}
```

,

- Create a document in `/properties/{propertyId}/sensors/Anemometry`:

```
`typescript

{

  type: "Anemometry",

  verificationStatus: "UNVERIFIED",

  hardwareType: "Ecowitt",

  value: 0

}
```

,

1. **Secret Generation:** Generate a unique password/secret token for the gateway and email it as a QR code to the user.

4. UI/UX Storefront Integration

The Shopify storefront theme will match Chainmail's core brand design system: **Cobalt & Gold / Slate & Gold**.

Theme Design System

- **Primary Background:** #030712 (Slate Black)
- **Secondary Cards:** rgba(17, 24, 39, 0.7) (Translucent Dark Blue)
- **Accents:** #D4AF37 (Metallic Gold)
- **Fonts:** Outfit (Headings) and Inter (Body)

Custom Liquid Elements

- **Network Node Counter:** A live ticker in the header showing: ACTIVE_NODES: 1,482 | TOTAL_COVERAGE_LOCKED: \$34,891,000`.
- **Map Twin Preview:** An interactive WebGL globe on the landing page showing glowing gold dots representing active coverage sensors in real time.
- **onboarding-tab widget:** Built-in section on the product page showing the simple "Plug, Scan, Insure" 3-step setup guide.

Client: Chainmail Solutions Inc. **Target Partner:** Shenzhen Fine Offset Electronics Co., Ltd.
Date: June 7, 2026 **Document Ref:** CM-HW-SPEC-2026-V1.0

1. Project Objective & Scope

Chainmail Solutions is launching a decentralized parametric insurance network. We are seeking an established OEM/ODM partner to manufacture, white-label, and factory-configure our weather sensor suite, branded as the **Chainmail Sentinel Series**.

The hardware must support direct cloud telemetry reporting, automated onboarding via factory-mapped serial numbers, and customized industrial casings.

2. Hardware Specifications & SKUs

We require white-labeled equivalents of the following standard Fine Offset / Ecowitt models:

SKU 1: Sentinel Gateway Hub (Model S-100)

Based on Ecowitt GW1100/GW1200

- **Core Function:** Receives RF telemetry from outdoor sensors and transmits normalized data over Wi-Fi (802.11 b/g/n) to the Chainmail ingestion API.
- **Internal Sensors:** Barometric Pressure, Temperature, Humidity.
- **Frequencies:** 915 MHz (US Market) and 868 MHz (EU Market).
- **Power:** 5V USB (Type-C preferred).

SKU 2: Sentinel CAT-7 Outdoor Sensor Array (Model S-700)

Based on Ecowitt Wittboy (WS90)

- **Core Function:** Solar-powered 7-in-1 outdoor weather station with built-in supercapacitor.
- **Sensors:** Ultrasonic Wind Speed, Wind Direction, Rain (piezoelectric rain sensor), Temperature, Humidity, UV, and Light.
- **Frequency Compatibility:** Paired with Model S-100 (915 MHz / 868 MHz).
- **Mounting:** Standard bracket for pole mount.

SKU 3: Sentinel Soil Saturation Probe (Model S-WH51)

Based on Ecowitt WH51

- **Core Function:** Wireless capacitive soil moisture sensor.
- **Frequency Compatibility:** 915 MHz / 868 MHz.

3. Firmware Customization & Telemetry Protocol

To support zero-configuration onboarding, gateways must be pre-flashed with custom default firmware settings at the factory.

Custom Server Configurations

The gateway firmware must be set with the following default values in the **Custom Server** settings:

Parameter	Configuration Value
Protocol Type	Ecowitt (HTTP POST, URL-encoded)
Server/IP	api.chainmailsolutions.ai

Parameter	Configuration Value
Path	/api/ingest/ecowitt
Port	443 (HTTPS / SSL Enabled)
Upload Interval	60 seconds
Default Password/Secret	Pre-flashed unique 32-character HMAC token mapped to MAC address

IMPORTANT

HTTPS Support:

Gateways must support secure TLS 1.2 or TLS 1.3 handshakes when posting data to our cloud API endpoint.

4. Branding & Industrial Design (White-Labeling)

We require the following structural modifications to the standard enclosures:

Logo Placement & Colors

- **S-100 Gateway:** Silk-screened **Chainmail Sentinel** logo (white colorway) centered on the top cover of the USB dongle casing.
- **S-700 Outdoor Array:** Engraved or weather-resistant silk-screened **Chainmail** shield emblem centered on the wind baffle fin.
- **Enclosure Color:** Dark Cobalt Blue (Pantone 295C) with Gold accent trim (Pantone 872C) representing the Chainmail corporate color scheme.

5. Out-of-Box Experience (OOBE) & Serial Mapping

To facilitate seamless network onboarding, the factory must generate and package custom insert cards:

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CHAINMAIL SENTINEL SYSTEM

1. Mount outdoor sensor array S-700 in clear view.
2. Insert S-100 Gateway into any USB power source.
3. Scan QR code below to bind device to your profile.

[QR CODE]

MAC Address: AA:BB:CC:11:22:33

Node Activation Key: [32-Character Unique Token]

Factory Logging Requirement

For every unit shipped, the manufacturer must provide a secure CSV inventory file containing:

1. Device MAC Address
2. Device Serial Number
3. Generated Activation Token
4. Printed QR Code Payload URL (e.g., <https://app.chainmailsolutions.ai/activate?mac=AABBCC112233&token=...>)

6. Business Terms & RFQ Questions

Please provide pricing and lead times based on the following volumes:

- **Initial Minimum Order Quantity (MOQ):** 1,000 units (Combo Kits containing 1x S-100 and 1x S-700).
- **Secondary Volume Pricing:** 5,000 units and 10,000 units.
- **Sample Units:** Lead time and cost for 5 custom-branded prototype samples (915 MHz).
- **Certification:** Confirmation of FCC (Part 15) and CE compliance for all RF transmitters.

RFQ Submission Instructions

Please submit technical specifications, material declarations, and unit price quotes in USD (FOB Shenzhen) to:

Greg Narcisi

Co-Founder, Chainmail Solutions Inc.

greg.narcisi@chainmailsolutions.ai`

,500 payout cap) just so we can get their sensor data and help protect the community.

The Claims Adjuster Agent: When a disaster hits, this AI looks at satellite photos to verify the damage. If it's confident, it pays the customer instantly. If it isn't sure, it sends the claim to a human to review—it is strictly programmed **never** to automatically deny a claim.

The Vigilance Oracle Agent: This AI constantly watches the weather channel. If it sees a Category 5 hurricane coming, it flags the map and readies our bank accounts to pay people before they even ask.

The Quant Brain Actuary Agent: This AI watches our bank account. If we take on too much risk in one state, it automatically buys "reinsurance" (backup insurance for us) to keep the company financially safe.

Your Day-to-Day Responsibilities

As a Product Owner, you will:

- **Monitor the Vision:** Look at the Enterprise Dashboard. Are the AI agents making decisions that align with the BallPark Initiative?
- **Prioritize the Backlog:** Work with the Product Manager to decide which neighborhood of our "city" needs improvement next.
- **Understand the Costs:** The Enterprise dashboard shows you exactly how many fractions of a cent the AI is spending to make decisions. You balance the cost of the AI against the value it provides.

You don't need to know how to code the AI. You just need to know what the AI **should** be doing to help our customers. Welcome to the team!

,500."*

Here is how you handle it:

- Explain that because their house is in an **Extreme Risk Zone** (like right on the beach), our Underwriting AI cannot offer them half a million dollars right now.
- However, explain that we **did not reject them**. We offered them the **IoT Baseline Policy**. This gives them ,500 of emergency cash, and in return, they install a weather sensor on their house. This helps us gather data to protect the rest of their neighborhood!

Need Help?

If the screen turns white, or the map stops loading, do not panic! Just take a screenshot and send it to your Product Manager with the Jira ticket label UI-BUG.

Welcome to Chainmail! We are glad you're here.

5). Require D-U-N-S number for corporate verification. **TestFlight / Internal Testing** T-4 Weeks Release to internal stakeholders via Apple TestFlight and Google Play Internal Testing. **Store Submission** T-2 Weeks Submit to Apple App Review and Google Play App Review. (Buffer time for initial rejection and resubmission). **Public Launch** T-0 App is live, indexed, and available for download.

2. Critical Apple App Store Requirements (iOS)

Apple is notoriously strict regarding financial and insurance applications. To pass the App Review Board, we must guarantee the following:

WARNING

digital settlement & Financial Regulations (Guideline 3.1.5)

Because Chainmail utilizes automated execution rules on Secure L2 Ledger L3 for payouts using \$USD-denominated digital settlement assets, the app **must** be submitted by a legal entity (not an individual developer) that is registered as a financial institution or has the appropriate licenses in the regions it operates.

- **Action Item:** Ensure our corporate D-U-N-S number is used for the Apple Developer Account, and our legal entity name exactly matches the app publisher name.

IMPORTANT

Data Privacy & Account Deletion (Guideline 5.1.1)

Apple absolutely mandates that any app allowing account creation **must** also provide an in-app option to completely delete the account and all associated data.

- **Action Item:** Build an automated "Delete Account" API route in the Vercel backend.

- **App Tracking Transparency (ATT):** If we use analytics that track across third-party apps, we must implement the ATT prompt.
- **Minimum Functionality:** The app cannot just be a "wrapped website" (Guideline 4.2). It must utilize native iOS features (e.g., Push Notifications, FaceID/TouchID for secure login, local caching).

3. Critical Google Play Requirements (Android)

Google Play has recently tightened restrictions on testing and financial apps.

WARNING

20-Tester Rule

New Google Play developer accounts must run a closed test with a minimum of 20 opted-in testers for at least 14 continuous days before you can apply for production access.

- **Action Item:** We must recruit 20 beta testers internally or via the Vigilance Grid network well before launch.

IMPORTANT

Financial Features Policy

Google requires a prominent "Financial Features Declaration" during submission. If the app facilitates insurance or digital settlement payouts, we must provide links to our regulatory compliance documentation and operating licenses in our Google Play Console declaration.

- **Data Safety Section:** We must accurately declare exactly what data is collected (e.g., location data from Vigilance Grid IoT integration, financial info) and whether it is encrypted in transit.

4. Next Steps for the Development Team

To transition our current Vercel web app into a compliant Native App:

1. **Architecture:** We must decide whether to wrap the Vercel app using a framework like Capacitor/Ionic, or rebuild the frontend natively using React Native.
2. **Privacy Policy URL:** A live, hosted Privacy Policy must be created on the public-facing website.
3. **Support URL:** A dedicated support and contact page must be established for user inquiries.

0/mo) post-funding for team access. **Database & Scaling** GCP Firestore / Blaze Plan 00.00 Scales dynamically; covers production compute and database queries. **Object Storage (Media/PII)** Cloudflare R2 \$0.00 Selected specifically to eliminate AWS/GCP data egress fees. **Geospatial &**

Convective Mapping XWeather API & Vexcel Imagery \$750.00 Commercial subscription for severe convective storm mapping and high-res post-storm imagery. **Core AI Routing & HITL Verification** Google Gemini 1.5 Flash & 3.1 Pro \$65.00 Production Gemini API tokens for IoT routing and human-in-the-loop validation. **Claims Fraud Audit** ISO Verisk ClaimSearch 50.00 Active double-claim checking API subscription. **Secure Registry & Smart Contract execution** Secure L2 Ledger Mainnet 5.00 Sponsored gas for transaction settlement. **Version Control** GitHub (Free Tier) \$0.00 Upgrade to GitHub Team (\$4/user) when engineering team expands. **Executive Compensation** COO Phase 1 (1099) \$0.00 (Equity/\$RMOR only) Transition to Full-Time Phase 1 (Salary + 8% Match) post-funding. **Total Monthly Burn ,090.00**

Future Capital Expenditure (Post-Funding)

Once institutional seed funding is secured, the following immediate upgrades will be unlocked to mature the ecosystem:

1. **CFAN Proprietary Licensing:** We will deprecate our reliance on open-source NWS/NCEP data and execute a commercial licensing agreement with the **Climate Forecast Applications Network (CFAN)**. This will drop our 5-day track error to ~131 miles, providing institutional credibility to reinsurers. *(Estimated: Negotiated Enterprise Contract)*
2. **Amazon IP Accelerator:** We will abandon the standard USPTO pending application process and utilize Amazon's vetted legal firms to fast-track our Sentinel IoT hardware Trademark and Brand Registry. *(Estimated: ,500 - \$3,000)*
3. **DVN Node Infrastructure:** Funding the initial liquidity pools on Secure L2 Ledger to incentivize legacy Third-Party Vendors (TPVs) to transition into distributed Validator Network (DVN) node operators.